

Shared evidence • Proposed water comparison

The water-trial planning board

Name _____ Date _____

Fictional teaching plan. An adult must adapt any real watering method to the plants and setting.

Question

How does daily water amount affect growth of these young bean seedlings over 10 days?

Keep comparable

Species, age, starting size, pot and compost; light, temperature, drainage and measurement times.

Care and honesty

Check daily. Keep actual records, including any missed measurement or necessary care change.

Groups

A1-A3: 5 mL each daily. B1-B3: 15 mL each daily. These amounts are a fictional teaching plan.

Record

On days 0, 3, 7 and 10: each plant's height in cm from soil to main shoot tip; leaf colour and condition.

W1. Name the change, measurement and two controls.

W2. Explain the difference between a prediction and a result.

Route 1 • Assemble a fair plan

Word bank: water amount / height / same / three / cm.

S1. Deliberately change...

- A. Water amount only
- B. Water, light and species

S2. Name two conditions to keep comparable.

S3. How many plants are planned in each group?

S4. Describe where the ruler starts and ends.

S5. Future results should be...

- A. Observed and recorded later.
- B. Invented from the prediction now.

Route 2 • Write the handover method

Give enough detail for another team to follow the proposed comparison.

M1. Write a testable investigation question.

M2. State the change and two controls.

M3. Write a short method including IDs, repeats, height rule and observation days.

M4. State one separate prediction and one care check.

Route 3 • Audit the weak plan

Weak plan: “Use one small bean plant in a dark cupboard with 5 mL daily, and one large pea plant in a sunny window with 15 mL. Measure once when convenient.”

T1. Identify three confounding differences.

T2. Repair the comparison using matched groups and repeats.

T3. Improve the observation schedule and measurement rule.

T4. A plant becomes severely stressed. What should the adult do, and how should this affect the record?

Exit and reflection

Point, draw, write or tell an adult.

E1. What factor is this plan changing?

E2. Why use three plants per group?

E3. Should a prediction be copied into the results as a measurement?

R1. Which idea did you improve or defend? What evidence helped?

R2. What useful question would you investigate next?
